

Effect of broccoli sprouts on insulin resistance in type 2 diabetic patients: a randomized double-blind clinical trial.



Use of antioxidant components is a new approach for improvement of insulin resistance (IR) as a main feature of type 2 diabetes and its complications. The aim of this study was to investigate the effects of broccoli sprouts powder (BSP) containing high concentration of sulforaphane on IR in type 2 diabetic patients. Eighty-one patients were randomly assigned to receive 10 g/d BSP (A, n = 27), 5 g/d BSP (B, n = 29) and placebo (C, n = 25) for 4 weeks. Fasting serum glucose and insulin concentration, glucose to insulin ratio and homoeostasis model assessment of IR (HOMA-IR) index were measured at baseline and again 4 weeks after treatment. Seventy-two patients completed the study and 63 were included in the analysis. After 4 weeks, consumption of 10 g/d BSP resulted in a significant decrease in serum insulin concentration and HOMA-IR ($p = 0.05$ for treatment effect). Therefore, broccoli sprouts may improve IR in type 2 diabetic patients.